

Experiment Name : 07

Experiment No : Study of transistor switching network using BJT

Objectives:

- To implement the switching circuit using BJT
- To observe the input and output waveforms of a BJT switching circuit

Theory:

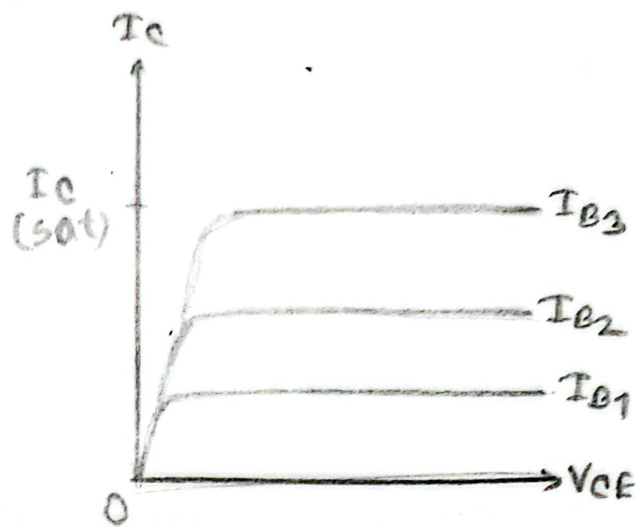
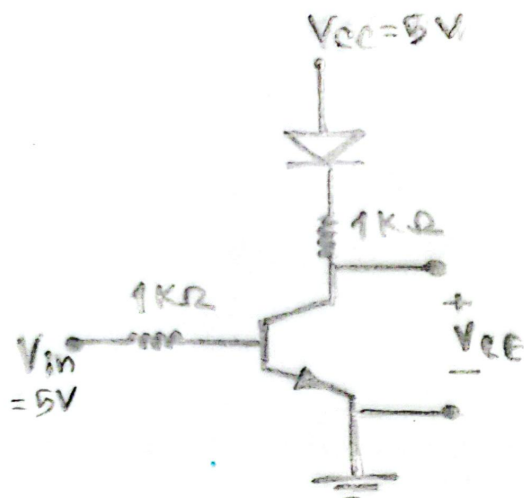
BJT is operated between cutoff and saturation states to act as a digital switch. A BJT switch is controlled by the base current (I_B).

- $I_B = 0 \rightarrow$ Transistor is in cutoff region $\Rightarrow$ No conduction $\Rightarrow$ open switch
- $I_B = \uparrow$ (sufficiently high) $\rightarrow$ Transistor enters saturation region $\rightarrow$ Conduction occurs $\rightarrow$ closed switch

Required Apparatus:

1. BJT (BD135, 1 piece)
2. Resistor ($1\text{ k}\Omega$, 2 pieces)
3. DC power supply (0-15V)
4. Project board
5. Multimeter
6. Connecting wires

Circuit Diagram:



Procedure:

- (i) The circuit was connected according to the figure.
- (ii) $V_{in} = V_B = 5V$ was provided and the output across V_{ce} was observed to be zero V.
- (iii) $V_{in} = V_B = 0V$ was provided and the output across V_{ce} was observed to be 5V.

(K)

Calculations:

When, $V_{in} = 5V$
 $V_{CE} = 0V$

When, $V_{in} = 0V$
 $V_{CE} = 5V$

Discussion and Conclusion:

In this experiment, the BJT successfully operated as a switch, turning ON in saturation with base input applied and OFF in cutoff when the base input was removed. The observed voltages and waveforms matched theoretical expectations, confirming the objectives of the experiment.

Precautions:

- All the connections should be connect
- Transistor ~~the~~ terminals must be identified properly
- Calibration must be performed properly